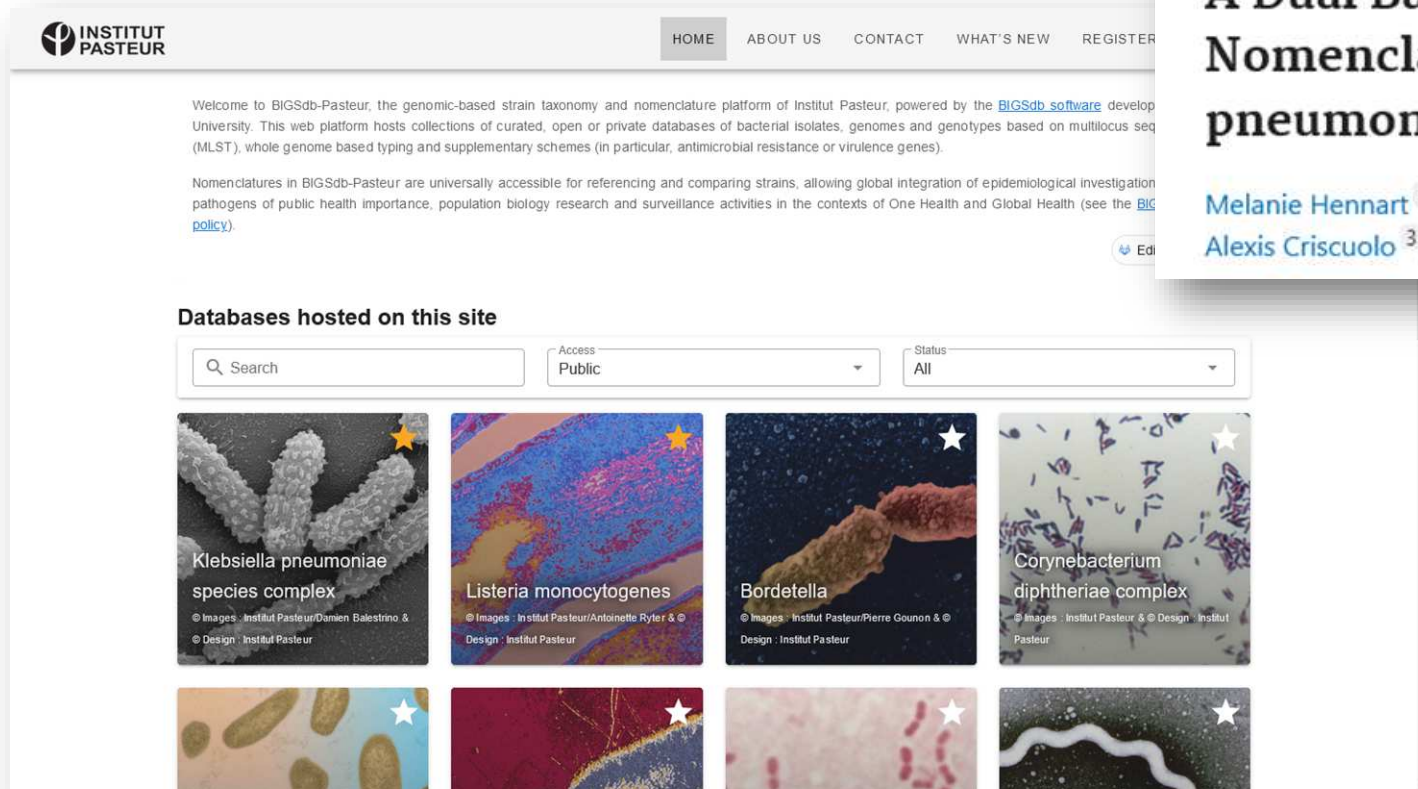


Life Identification Numbers (LIN) using cgMLST provide a stable nomenclature



> [Mol Biol Evol.](#) 2022 Jul 2;39(7):msac135. doi: 10.1093/molbev/msac135.

A Dual Barcoding Approach to Bacterial Strain Nomenclature: Genomic Taxonomy of *Klebsiella pneumoniae* Strains

Melanie Hennart ^{1, 2}, Julien Guglielmini ³, Sébastien Bridel ¹, Martin C J Maiden ⁴, Keith A Jolley ⁴, Alexis Criscuolo ³, Sylvain Brisse ¹



Life Identification Numbers (LIN) using cgMLST provide a stable nomenclature

Genome sequences ↓ cgMLST profiles ↓ cgST ↓		Bin number:	1	2	3	4	5	6	7	8	9	10
		Max. allelic similarity*:	19	44	439	586	619	622	625	627	628	629
		Min. allelic difference*:	610	585	190	43	10	7	4	2	1	0
		Bins left thresholds:										
		Closest genome (similarity %)	0	3.02	6.99	69.79	93.16	98.41	98.88	99.36	99.68	99.84
Genome A	Initialization		0	0	0	0	0	0	0	0	0	0
Genome B	A (3.50%)		0	1	0	0	0	0	0	0	0	0
Genome C	B (99.0%)		0	1	0	0	0	0	1	0	0	0
Genome D	B (7.00%)		0	1	1	0	0	0	0	0	0	0
...	...		...									
Genome X	Y (5.00%)		0	2	0	0	0	0	0	0	0	0
Genome Z	X (98.90%)		0	2	0	0	0	0	1	0	0	0

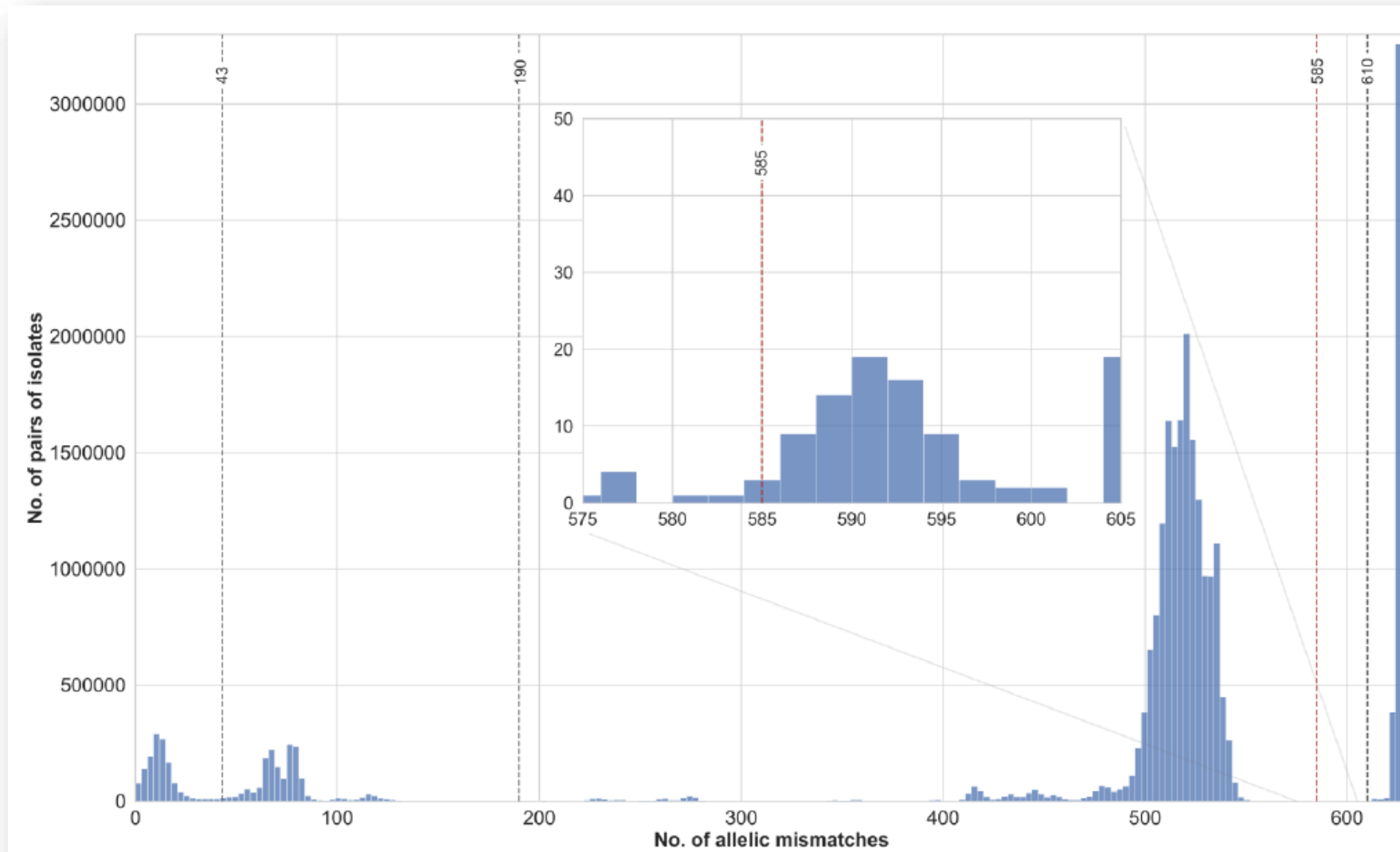
Coding steps (genome Z):

(i) (ii) (iii)

Palma *et al.* 2024. Bacterial strain nomenclature in the genomic era: Life Identification Numbers using a gene-by-gene approach. bioRxiv 2024.03.11.584534



Plotting the distribution of pairwise distances for a diverse set of 7000 *K. pneumoniae* assemblies indicates discontinuities reflecting the population structure



Hennart *et al.* 2022. A Dual Barcoding Approach to Bacterial Strain Nomenclature: Genomic Taxonomy of *Klebsiella pneumoniae* Strains *Mol Biol Evol* **39**: msac135



LIN code prefixes provide nomenclature for specific branches within the population structure

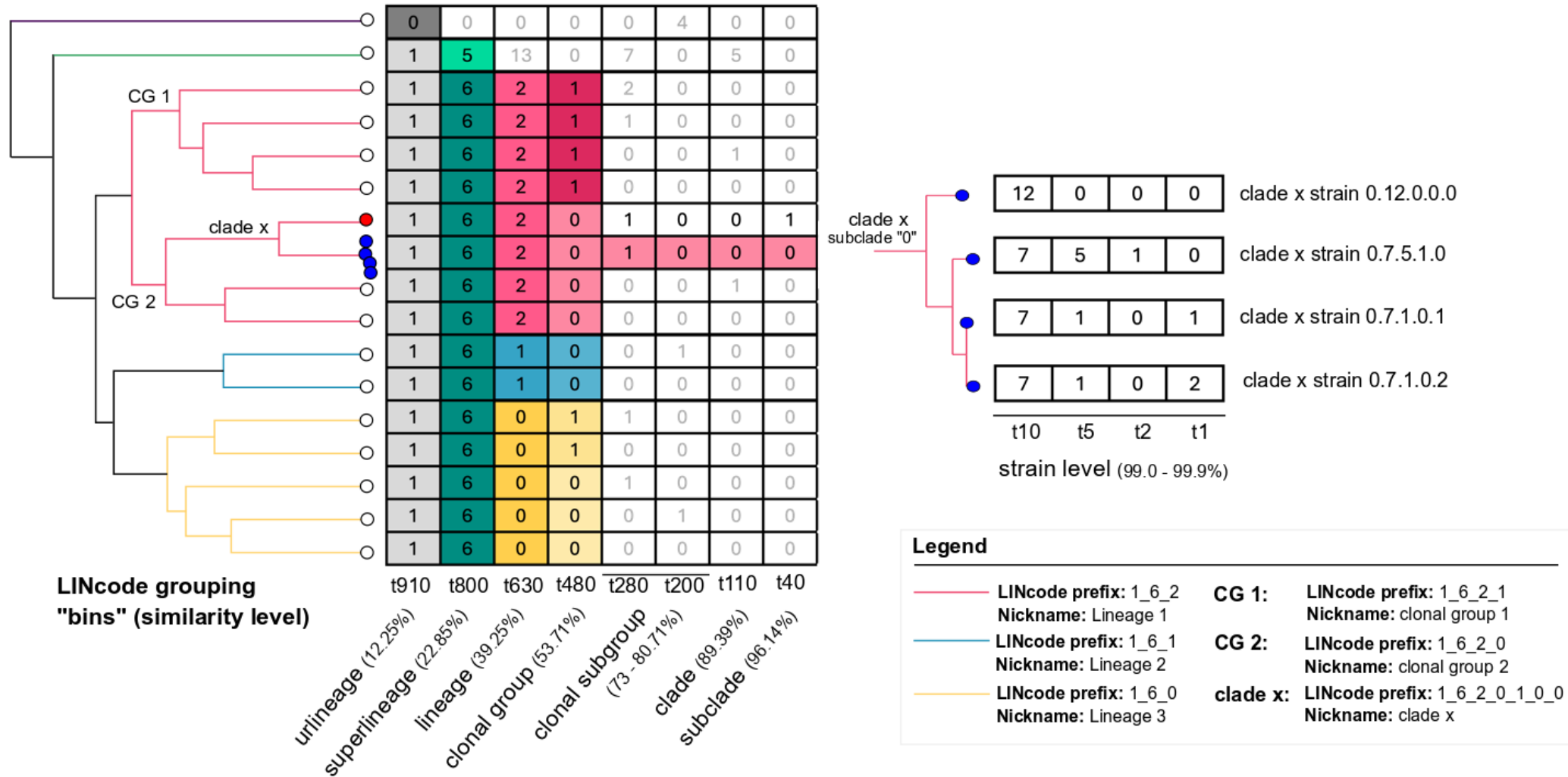


Figure provided by
Made Krishna
(unpublished)

We can now use LIN codes to identify closest matching genomes following a sequence query

cgMLST v1



Matching profiles

Closest profile: **cgST-1448**

Mismatches: 164

Loci matched: 873/1037 (84.2%)



Similar profiles (determined by LIN codes)

Partial LIN code: 0_1_0_2_0_0

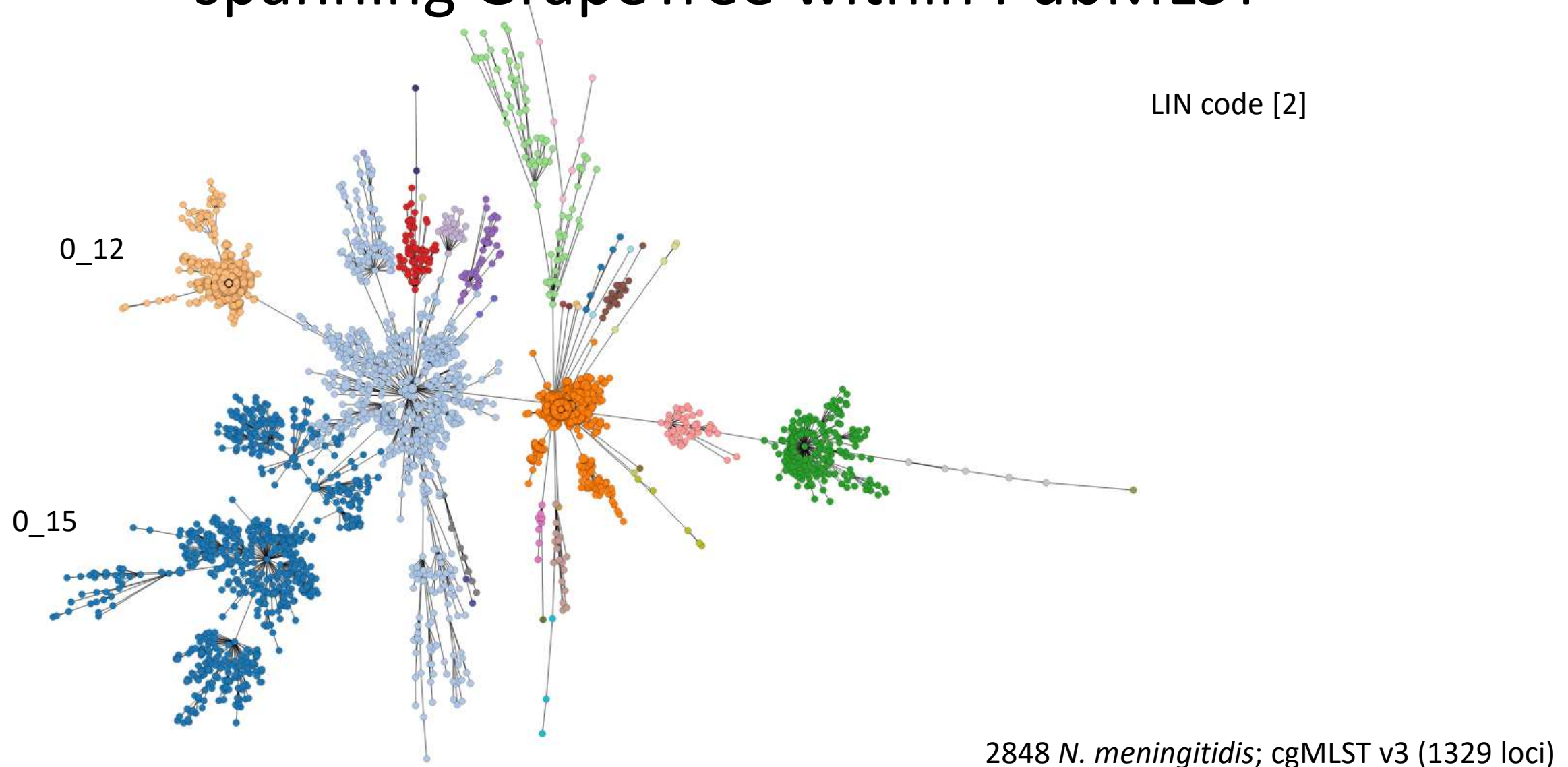
Show all thresholds

Prefix	Threshold	Matching profiles	Isolates
0_1	800	1960	PubMLST isolates 663
0_1_0	630	1153	PubMLST isolates 406
0_1_0_2	480	414	PubMLST isolates 141
0_1_0_2_0	280	62	PubMLST isolates 25
0_1_0_2_0_0	200	62	PubMLST isolates 25

LIN code schemes have been developed for a range of pathogens and implemented on PubMLST or BIGSdb Pasteur

- *Klebsiella pneumoniae*
- *Corynebacterium diphtheriae*
- *Haemophilus influenzae*
- *Streptococcus pneumoniae*
- *Staphylococcus aureus*
- *Moraxella catarrhalis* (in development)
- *Neisseria gonorrhoeae*
- *Neisseria meningitidis* (in development)
- *Pasteurella multocida* (in development)

LIN code prefixes can be overlaid on a minimum-spanning GrapeTree within PubMLST



LIN code prefixes can be overlaid on a minimum-spanning GrapeTree within PubMLST

LIN code [3]



2848 *N. meningitidis*; cgMLST v3 (1329 loci)

LIN code prefixes can be overlaid on a minimum-spanning GrapeTree within PubMLST



2848 *N. meningitidis*; cgMLST v3 (1329 loci)